Royal Devon University Healthcare NHS Foundation Trust

Clinical Guideline for:
Sedative pre-medication for children and adults under the care of an Anaesthetist

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SUMMARY

This guideline outlines the process for pre-procedure sedation for paediatric and adult patients undergoing general anaesthesia under the care of an anaesthetist at RDUH (Eastern and Northern)

KEY POINTS

1. This guideline is intended for the sedative pre-medication of children and adults, who are **under the care of an anaesthetist only**. This should not be used in circumstances where an anaesthetist is not involved in the patient's care, when other guidance is available e.g. Bramble Ward paediatric sedation protocol which covers sedation for children under the care of the paediatric team.

2. This guideline will ensure the safe prescribing and administration of sedative drugs to children and adults with learning disabilities and challenging behaviour.

3. The guideline will ensure that the administration of sedative drugs used before general anaesthesia are given in line with current legislation in regard to capacity and consent.

4. The guideline applies to Anaesthetists and Nursing Staff involved in preparing patients for General Anaesthesia.

5. The essential elements of this guideline are:

    - Midazolam (oral or buccal) is first line
    - Alternatives to midazolam may be required in certain circumstances
    - Combinations of 2 (or more) medications may occasionally be required
    - In some children, the recommended doses per kilogram and maximum doses can be exceeded (above that in Appendix 1 and in the BNF/ BNFC). This is the responsibility of the prescribing anaesthetist, based on clinical experience.

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1. INTRODUCTION

1.1 This clinical guideline is designed to provide advice, guidance and direction to staff whilst leaving room for professional judgement, and adaptation, to fit individual circumstances.

1.2 This purpose of this document is to formalise current practice and improve the prescribing and administration of sedative pre- medication when used before general anaesthesia.

2. BACKGROUND

2.1 Sedation prior to an anaesthetic may be required for many reasons including anxiolysis and to improve the patient's experience of anaesthesia and surgery. Groups of children and adults who may particularly benefit include those with additional needs (e.g. patients with autism or neurodevelopmental conditions, learning disability or emotional dysregulation), children having major surgery, repeated procedures, or those who have had a previous adverse experience of the hospital environment.

2.2 The advantages of sedation are that it can reduce both patient and carer anxiety therefore improving the overall experience. It can provide amnesia and may reduce behavioural changes post operatively.

2.3 The advantages must outweigh possible disadvantages including worsening anxiety associated with administration of the medication, paradoxical reactions that can occasionally be seen (e.g. with benzodiazepines) and the potentiation of sedative side effects of drugs such as opioids.

2.4 Sedation is one method of reducing anxiety but non-pharmacological methods should also be considered and may be used in addition to or as well as medications. Play Specialists and the adult Learning Disability teams are particularly helpful in this regard and a formal referral can be made to them either ahead of admission or on the day of surgery for paediatric and adult patients respectively.

2.5 On the Northern Site these patients will have a "Reasonable Adjustment Pathway" completed prior to surgery. On the Eastern site a 'This is me' passport will be introduced for autistic children and patients with ADHD or chronic illness which will help to describe what particularly triggers anxiety and what helps to reduce distress. A hospital passport is also available for adults with additional needs on the Eastern side.

2.6 This guideline has been developed to provide clarity on selecting sedative medication and managing children or adults once the sedative medication has been administered.

2.7 **These guidelines are intended for patients requiring pre-operative or pre-anaesthetic sedation under the care of an anaesthetist only.**

These guidelines should not be used for sedation for other procedures on Bramble ward or Caroline Thorpe Wards, nor any adult wards or emergency departments when not under the direct care of an anaesthetist – in this circumstance, please refer to relevant guidelines available on the Hub such as Bramble paediatric sedation protocol or adult rapid tranquilisation guidelines as appropriate for the clinical situation, or seek guidance from senior medical staff.

2.8 Adults with very complex requirements who may need sedation or anaesthesia at home or in the hospital car park are highly unusual and require extensive MDT planning before the day of the procedure. These patients are outside the remit of this guideline and should be referred to the adult Learning Disability team who can co-ordinate the pre-procedure discussions, meetings and planning.

3. DEFINITIONS

3.1 Learning Disability

- According to the World Health Organisation ICD-10, a learning disability is a condition of arrested or incomplete development of the mind, with mild to profound impairment in cognition, language, motor and social skills.
- People with a learning disability may take more time to understand complex information and co-operate with others
- The level of care depends upon individual factors, including both the severity and the nature of their disability.

3.2 Sedation

Sedation is a drug-induced depression of consciousness, which is a continuum culminating in general anaesthesia. The Academy of Medical Royal Colleges (AoMRC) and the American Society of Anaesthesiologists (ASA) define 4 levels of sedation:

- Minimal sedation (anxiolysis) – normal response to verbal stimulation
- Moderate sedation – purposeful response to verbal or tactile stimulation
- Deep sedation – purposeful response only to repeated or painful stimulation
- General anaesthesia – no response even with painful stimulus

As patients pass along the continuum from minimal to deeper levels of sedation, increasing depression of normal physiological responses occurs.

4. MAIN BODY OF GUIDELINE

SEDATIVE PRE-MEDICATION

4.1 Aims of Pre-Medication

The aims of premedication are to reduce patient and carer anxiety and improve overall experience. It may provide amnesia and also could reduce behavioural changes post operatively. The anticipated effects of the sedation should be explained to the parents/ carers (and child if appropriate). If the patient has received pre-medication previously, it is useful to determine the medication used, route, effectiveness and any adverse events that may have occurred. The advantages must outweigh the potential disadvantages (such as worsening anxiety associated with the administration of the medication or paradoxical reactions that can be seen).

The medication and route of administration chosen will depend on the particular circumstances on the day, the patient's previous experiences with sedative pre-medication and co-operation and tolerability of each possible route. The fasting status should be confirmed before any sedative medication is given.

4.2 Patients who may benefit from sedative pre-medication:

- Children or adults with special educational or behavioural needs
- Children having major surgery
- Children having repeated procedures
- Children who have had a previous bad experience of the hospital environment
- Children or adults with anxiety
- Children with past history of stormy anaesthetic induction or emergence

4.3 Contra-indications to sedative pre-medication include:

- Difficult airway
- Increased aspiration risk
- Severe obstructive sleep apnoea
- Raised intracranial pressure
- Severe hepatic or renal impairment
- Previous allergic or behavioural reaction to the premedication planned

- New or unexplained reduced saturations on room air
- Severe respiratory disease e.g. COPD requiring long term oxygen therapy
- Acute systemic illness e.g. severe sepsis

In these situations, if a sedative premedication is warranted, please discuss with a paediatric anaesthetist or on call anaesthetic consultant in the first instance. The decision to give sedative pre-medication will need to be made on an individualised basis.

4.4 Roles and Responsibilities

4.4.1 The Anaesthetist is responsible for:

- Choice of sedative premedication
- Ensuring that sedative premedication is prescribed correctly on the appropriate prescription chart, and where possible in line with the current guideline.
- Ensuring that the patient is in a safe environment
- Ensuring parents or carers are aware of the effects of any sedation administered.

4.4.2 The designated nurse is responsible for:

- Checking the prescription and preparing the sedative premedication
- Ensuring that the premedication is safely and effectively administered
- Communicating directly with the prescribing Anaesthetist if there are any problems with the prescription or administration of the medication(s)
- Maintaining a safe environment for the patient once sedation is administered which will include appropriate positioning and monitoring to avoid falls and unrecognised hypoxia (see section 4.6)

4.5 Consent

- Where sedation is to be administered, informed consent must be gained
- Where there is reason to doubt that the adult concerned has capacity to consent, a formal mental capacity assessment must be undertaken and recorded in the notes.
- If the adult is assessed as not having capacity, then a decision should be made in their best interests. This should be the least restrictive option with reference given to the RDUH Consent Policy

- If preoperative oral or intramuscular sedation is planned then a discussion must occur with the patient (if appropriate) and/or the parents and carers beforehand. This discussion must cover the indication for premedication, the risks and benefits and in case of failure the plan for ongoing management of the patient. This discussion must be documented.
- All children should be included in discussions regarding their health and treatment as much as possible given their level of comprehension.

4.6 Choice of medication

4.6.1 Choice of medication will depend on the particular circumstances at the time and includes:

- Previous experiences with premedication
- Formulation
- Pharmacology and contra-indications
- Tolerability, preferred route and their co-operation

4.6.2 Combinations of medications can be used in certain circumstances (e.g. previous failure of a single agent alone) such as oral midazolam and clonidine together.

4.6.3 In some patients with extremely challenging behaviour, up to three different medications may be required under the direction of an anaesthetist. (See recommendations for 2nd, 3rd and 4th line options in these instances).

4.6.4 **In some children and adults, the recommended doses per kilogram and maximum doses can be exceeded (above that in Appendix 1 and in the BNF/ BNFC)**
This is the responsibility of the prescribing anaesthetist and will be based on experienced clinical judgement.

Most commonly this may be necessary with buccal midazolam as the doses in the BNF for ‘conscious sedation’ may be lower than those required for clinical effect ahead of anaesthesia. There is considerable evidence of safety of the 300mcg/kg dose as an anaesthetic pre-medication and it is recommended in the anaesthetic literature.
This will be prescribed and requested by the consultant anaesthetist, clearly documented and communicated with the nursing team involved.

4.6.5 Pathway for choosing sedative pre-medication (see Appendices 1 and 2 for tables giving dosing information) for children and adults with learning disabilities.

1st line	• Midazolam (oral or buccal) •Consider clonidine (oral) OR dexmedetomidine (oral, buccal or intranasal) instead if midazolam has had excitatory effects previously
2nd line	• Midazolam (oral or buccal) • PLUS • Clonidine (oral) OR Dexmedetomidine (oral, nasal or buccal)
3rd line	• Midazolam OR clonidine/ dexmedetomidine • PLUS • Ketamine
4th line	• **Discuss with consultant anaesthetist** • 3 agents OR IM ketamine

4.6.6 If combinations of medications are required (i.e. 2nd, 3rd or 4th line), consideration should be given to an anaesthetic trainee, practitioner or consultant remaining with the child (or adult) or being on the ward continuously unless there is previous experience of this combination of medications being used safely in that patient. This should particularly be considered if the child is in a location distant to theatres e.g. Caroline Thorpe or Bramble ward and cannot have 1:1 nursing care. This should be mandatory if 3 agents are used.

4.6.7 Either clonidine OR dexmedetomidine should be given, NOT both.

4.6.8 The oral preparations may be added to a small volume of flavoured squash to help with any unpleasant taste i.e. dilute with an equal volume of squash. Medications given orally can be mixed in the same drink e.g. midazolam plus clonidine/ dexmedetomidine, midazolam plus ketamine.

4.6.9 If intramuscular ketamine is required, it should be discussed with parents/ carers and staff in advance. It requires 3-5 minutes to work and will need sufficient staff, appropriate monitoring and equipment and the ability to transfer the patient up to theatre rapidly. See Appendix 4.

4.6.10 If the patient refuses, spits out or vomits the medication, please contact the prescribing Anaesthetist as soon as possible. Do not transfer the patient up to theatre without discussing with the anaesthetist first.

4.6.11 Care should be taken when prescribing in the overweight child. The SOBA guideline 'Anaesthesia for children living with obesity' gives guidance on dose adjustments based on BMI centiles for pre-medication

4.6.12 For adults who need a pre-medication for anxiety reasons only (ie those who do not have a learning disability or only a very mild one) first line pre-medication is a benzodiazepine such as temazepam or diazepam.

4.7 After administration

- The patient should remain on the ward under the direct supervision of a responsible adult such as a parent or guardian.
- When they become drowsy, they should lie in bed or on a trolley.
- If they fall asleep, oxygen saturations should be continuously monitored using pulse oximetry.
- Oxygen should not routinely be applied.
- If pre-medication is inadequate, additional medication may be required, at the discretion of the prescribing anaesthetist.

Appendix 1

Sedative Premedication Dosing in Paediatric Patients

(Note: every option of medication shown below will not always be available in every location which is why multiple different options are shown).

Drug	Route	Formulation	Dose	Onset of action	Notes
Clonidine	Oral	(use IV solution orally) 150 micrograms/ml or Tablets 25 or 100 micrograms	4 micrograms/kg Max 200 micrograms	45 mins	Tasteless
Dexmedetomidine	Intra nasal	100 micrograms/ml solution for injection may be administered via these routes – this is not within the product license	2-3 micrograms/kg rounded up to nearest 5 micrograms Max 200 micrograms	25 minutes	Tasteless Use MAD device to deliver drug May have prolonged recovery due to duration of action Bradycardia is common
	Buccal		1-4 micrograms /kg rounded up to nearest 5 micrograms Max 200 micrograms	25 minutes	
	Oral		4 micrograms /kg	30 - 45 minutes	
Midazolam	**Oral/ enteral** Miprosed Amsed	 5mg/ml 2.5mg/ml	0.25-0.5 mg/kg Max 20 mg	30 mins	May see paradoxical agitation
	Oral	IV solution administered orally (5mg/ml) NOTE 10mg/2ml midazolam not available at RDUH East theatres)	0.25-0.5 mg/kg Max 20mg	30 mins	Very bitter taste
	Buccal Epistatus Buccolam	 10mg/ml 5mg/ml NOTE different concentrations	300micrograms/kg Maximum per dose: 1-4yrs = 5mg 5-9 yrs = 7.5mg 10-17 yrs = 10mg	10-20 mins	Note these are the status epilepticus doses – as discussed and agreed by paeds pharmacist
	Intranasal Midazolam		0.2mg/kg	5-10mins	

		Midazolam 40mg/ml and lignocaine 20mg/ml	Max 10mg midazolam (=0.25ml) Can give second dose if first ineffective		Use MAD device to administer medication
Ketamine	Oral	50mg/ml or solution for injection may be administered via these routes NOTE different concentrations (Outside product license)	5-10 mg/kg Maximum 400mg	20-30 minutes	
	Intra muscular		5mg/kg Maximum 400 mg	5 minutes	
Ketamine with Midazolam	Oral		Ketamine 3mg/kg plus Midazolam as above	20-30 mins	Over 1 year old only

Notes

- The 100 microgram Clonidine tablets can be crushed and dispersed in water. This can be considered to minimise the volume of liquid.
- Dexmedetomidine is shorter acting than clonidine, has little if any respiratory depression, is easy to administer intranasally and is not as irritant as intranasal midazolam.
- Oral and buccal midazolam can be increased above the recommended maximum dose but this is the sole responsibility of the Anaesthetist concerned.
- Ketamine can be mixed and given with midazolam at the dose described above.
- Dexmedetomidine can be mixed and given with midazolam at the doses described above.
- Ketamine is a good analgesic but is hallucinogenic and patients should be recovered in a relatively quiet and non-stimulating environment.
- If asked to prescribe a premedication in order to facilitate attendance for surgery. This is not a usual process and should undergo an MDT discussion prior to consideration. Alternative strategies should be sought, such as desensitisation.

- Intranasal midazolam (with lidocaine to reduce stinging) is reserved for children or adults with very challenging behaviour and is not first line. It has been used by dental services in the Trust very effectively to allow treatment. There is extensive evidence that IN midazolam can be safely used in children of all ages.

Appendix 2

Sedative Premedication Dosing in Adult Patients

(Note: every option of medication shown below will not always be available in every location which is why multiple different options are shown).

Drug	Route	Formulation	Dose	Onset of action	Notes
Clonidine	Oral	(IV solution given orally) 150 micrograms/ml Or Tablets 25 or 100micrograms	4micrograms/kg Max 200micrograms	45-60 mins	
Dexmedetomidine	Intranasal	100 micrograms/ml Outside of product license	2-3 micrograms/kg Max 200 micrograms	25 mins	
	Buccal		1-4 micrograms/kg Max 200 micrograms	25 mins	
	Oral		2-4 micrograms/kg	20-45 mins	
Midazolam	Oral/ enteral CHECK FORMULATION AND CONCENTRATION	10mg/ml (Epistatus) – given ORALLY 5mg/ml (Buccolam oromucosal solution – given ORALLY)	0.25-0.5mg/kg Max 20mg	20-30mins	

		Amsed oral solution 2.5mg/ml Miprosed oral solution 5mg/ml IV solution (5mg/ml) can be given orally (Note 10mg/2ml midazolam not available at RDUH East in theatres or on Knapp ward)			
	Buccal CHECK CONCENTRATION	**Epistatus** 10mg/ml **Buccolam** 5mg/ml	0.3 mg/ kg Max 10 mg	20 mins	
	Intranasal Midazolam	Midazolam 40mg/ml and lignocaine 20mg/ml	0.2mg/kg Max = 10mg midazolam (=0.25ml) Can give second dose if first ineffective	5-10mins	Use MAD device to administer medication
Temazepam	Oral	Tablets	10-30mg	1-2 hours	Useful in anxious adults Half-life 8-22hours
Diazepam	Oral	Tablets	2.5-10mg	1-2 hours	Useful in anxious adults Half life 20-200 hours (active metabolites)
Lormetazepam	Oral	Tablets	1-2mg		1mg in elderly/ frail or <50kg Half-life 10-12 hours

Lorazepam	Oral	Tablets	0.5-1mg		Half-life 10-20 hours
Ketamine	Oral	50mg/ml solution for injection (can be given orally) Outside of license	5mg/ kg	20-30 mins	
	Intramuscular		5mg/kg	5 mins	

Notes

Adults requiring simple anxiolysis:

- For adult patients who require anxiolysis before surgery, clear explanation and calm dialogue will often suffice to alleviate their fears
- In those whom pre-medication is indicated for anxiety alone, benzodiazepines remain the mainstay. The difference between these drugs is marginal.
- Consider lower doses of benzodiazepines in the elderly or frail patients.
- Taking into consideration speed of onset, duration of action, side-effects and quality of anxiolysis, the most appropriate sedative premed is:
  - **Lormetazapam: 1-2 mg (1mg in elderly or <50kg) or Lorazepam: 0.5-1mg**
    - (Note that the onset of action of lorazepam is longer – 90mins vs 20 mins for lormetazapam, and has a longer duration of action). See supporting information below.

SUPPORTING EVIDENCE

- Lormetazepam undergoes hepatic metabolism with nearly constant kinetics even if repeatedly administered. It is safe in patients with renal impairment but used with caution in those with hepatic impairment. Lormetazepam has improved 'controllability' compared to midazolam. This may lead to less frequent agitation and over-sedation.
- Lormetazepam has some advantages over diazepam with a greater anxiolytic effect and less hangover/ side effects.
- Lorazepam has a significantly longer onset 40 mins versus 10 mins for lormetazepam although the anxiolytic effect may be less effective with lormetazepam.

Adults with learning disabilities:

In patients with more severe agitation or those with learning difficulties or special needs, lorazepam/ lormetazepam may well be insufficient, and po/ buccal midazolam may be a significantly better option.

There have been isolated cases of significant respiratory depression from using buccal midazolam in adult cases. If buccal midazolam is required, then an anaesthetist must be available to attend to the patient. In difficult cases, pre-operative planning for the use of midazolam and/or other agents will have been made. The anaesthetist should be present on the ward until transfer to theatre for these cases.

Appendix 3: Sedative Premedication Flow Chart

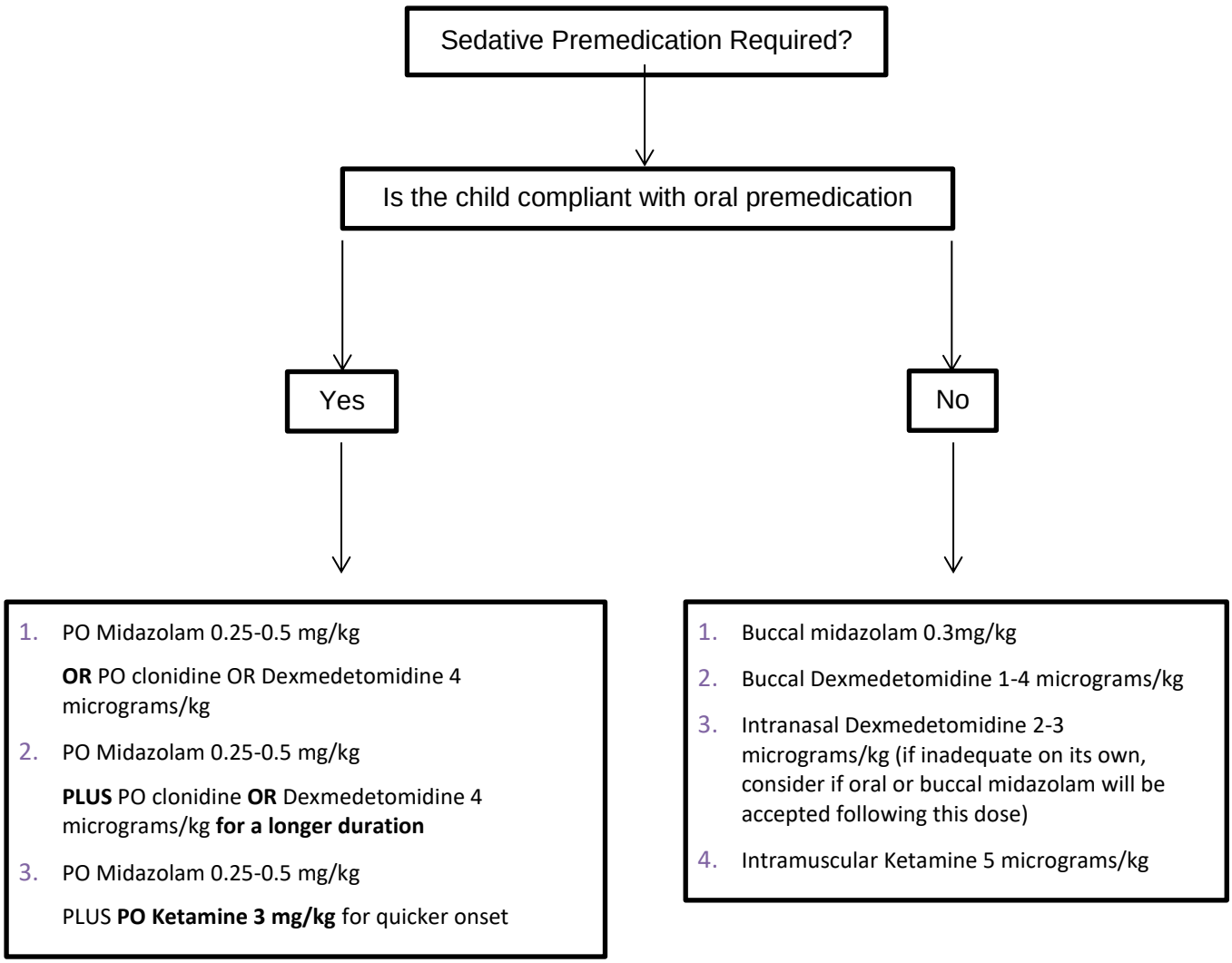
Advantages	Disadvantages
• Reduces both patient and parental anxiety and improves overall satisfaction with the experience • Provides ante grade amnesia • Reduces postoperative behavioural changes and adverse outcomes in children	• Noncompliance by the child may worsen anxiety • Paradoxical reaction may be seen in response to some drugs e.g. Midazolam • May potentiate the effect of other sedative drugs
Indications	**Contraindications**
Need for sedative premedication should be part of pre-anaesthetic assessment of any child or adult with special needs. The following patients are likely to benefit from premedication. • Children or adults with special educational or behavioural needs • Children having major surgery • Children having repeated procedures • Children who have had a previous bad experience of the hospital environment • Children or adults with anxiety • Children with past history of stormy anaesthetic induction or emergence	The following are some conditions where a careful risk assessment must be made prior to prescribing sedative premedication • Anticipated difficult airway • Increased aspiration risk • Obstructive sleep apnoea • Raised intracranial pressure • Acute systemic illness e.g. severe sepsis • Reduced SpO2 on room air • Severe renal or hepatic impairment • Previous allergic or behavioural reaction to the premedication proposed

Practical considerations
• **Confirm fasting status** (refer to fasting guidelines for all patients having an anaesthetic at RDUH) • **Discuss with the parents or carers of the child/patient** (where appropriate) **and document** what they can expect from the premedication • **Prescribe the premedication** on the drug / anaesthetic chart, including **route dose and time** when it should be administered (different drugs have variable onset times see chart below) • **Inform the preoperative / day surgery nurse** responsible for the patient that a "premed" has been prescribed to ensure that it is given at the appropriate time. • Oral premedication drugs can be given **mixed with a small volume** (5-30ml) **of clear solution** (e.g. ribena, apple juice or paracetamol mixture) which the child usually accepts to mask the unpleasant taste • Sedative drugs should be provided in a **safe environment** where resuscitation equipment and drugs can be accessed • A sedated child / adult should be **appropriately monitored** at all times.

Tips and trouble shooting

- Oral midazolam has a quick and reliable onset and is most frequently used first line agent.
- There is some evidence that Miprosed is more tolerable than the traditional IV midazolam or Amsed
- Oral midazolam may be combined by the anaesthetist with dexmedetomidine or clonidine for a longer duration of action.
- Ketamine is hallucinogenic and so patients should be recovered in a relatively quiet non stimulating environment.
- Beware additive sedative effects and consider dose reduction if using more than one medication at the same time.
- In children with previous **emergence agitation** or **paradoxical reaction to midazolam**, clonidine or dexmedetomidine given either as a premedication or intraoperatively, may have a beneficial effect.
- Intranasal /transmucosal (buccal) route often has a quicker onset and may be used when oral premedications are refused, keeping in mind that trying the intranasal route in a distressed child may pose a risk of injury to staff and child (beware that absorption from the intranasal route may be variable)
- Use a MAD (mucosal atomizing device) to deliver the drug effectively and safely when using the intranasal route- see appendix 5
- Intra muscular drugs (Ketamine) may be used in exceptional circumstances as an alternative in noncompliant patients if premedication is felt necessary (see appendix 4)
- It is not normal practice that a sedative is prescribed prior to coming to hospital. Alternatives should be sought (such as desensitization) and an MDT should be undertaken so that shared decision making can be undertaken in the best interest of the patient.

Suggested Pathway for Choosing Sedative Premedication in Children



Appendix 4

Ketamine Intramuscular Injection

- Ketamine may be administered by intramuscular injection to patients in whom administration of oral premedication has failed.
- Intramuscular ketamine has a quicker onset of action (seconds to minutes).
- The administration of intramuscular ketamine must be discussed with the child/parent/carer and this discussion should be documented.
- The discussion should include alternative courses of action, a plan for administration, and the immediate effects of ketamine.
- Prepare the “Ketamine intramuscular injection” to a dose of 5mg/kg (Max 400mg for paediatric patients) in a 5ml syringe
- The parents/carers should hug or distract the child, ideally seated or lying down and consideration should be given to the use of the “Sensory Unit” and play specialists.
- Ketamine can be effectively administered in one of the side rooms on the day surgery unit or in the anaesthetic room.
- Keep the patient’s shoulder (deltoid), thigh (vastus lateralis) or buttock (ventrogluteal) exposed if possible.

Appendix 5:

Mucosal Atomizer Device (MAD) and intranasal administration of dexmedetomidine

- Use the MAD device to administer 2-3 micrograms/kg of Dexmedetomidine neat (100 micrograms/ml)

- Intranasal dexmedetomidine may be challenging to deliver to patients who are already anxious and unable to tolerate oral medications. The administration of intranasal dexmedetomidine must be discussed beforehand with the child/parents/carers and team and this discussion must be documented. The discussion should include alternative courses of action and a plan for administration.

- Use of concentrated medication solutions in a small volume (0.2– 0.3 mL per nostril) is considered ideal, with volumes in excess of 1 mL per nostril not being reliably absorbed as a result of mucosal surface saturation and run-off from the nasal cavity

- Dexmedetomidine is 100mcg/ml solution

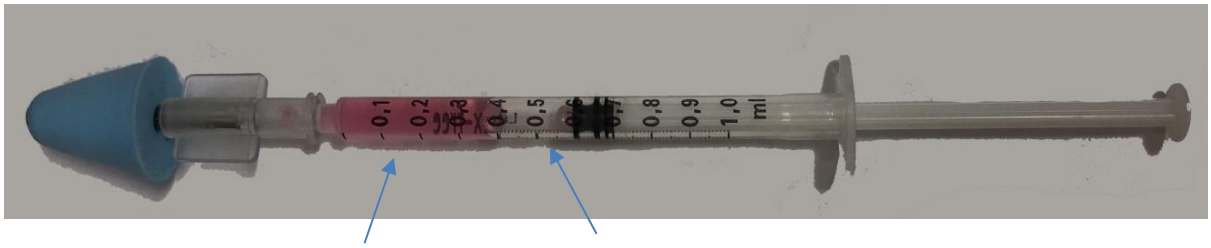
    - For doses from 10 - 79 micrograms: dilute required dose to a final volume of 0.8mL with sodium chloride 0.9% and give 0.4mL into each nostril.

    - For doses from 80-200 micrograms use without further dilution, dividing the total dose between the two nostrils.

- A 2ml syringe is generally the right size to use, and make sure that the sponge is gently but firmly nestled in the nostril to avoid shooting the device across the room

- The internal volume of the atomiser creates a “dead space” which has the potential to retain a proportion of the drug intended for administration and cause incomplete delivery of the intended drug dose. The exact volume of the dead space will depend on both the specific MAD and syringe type used, but is typically in the region of 0.1-0.16mL. Details of the volume for a specific MAD can usually be found in the package leaflet provided with the product.

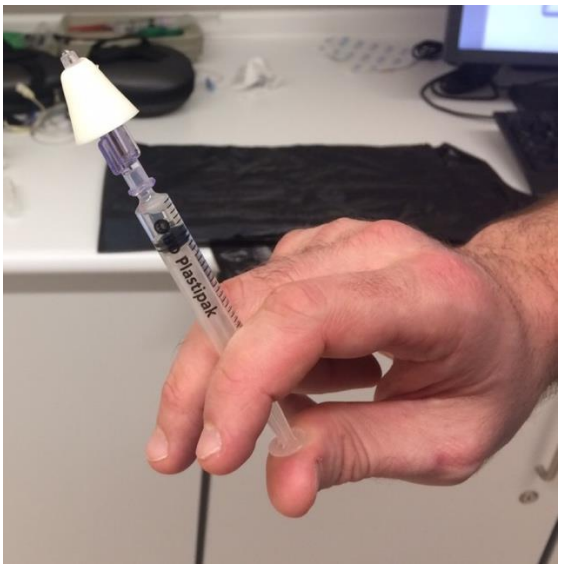
- One method for ensuring that the dead space does not lead to delivery of a reduced dose is to use an air bubble introduced into the syringe before administration. The syringe should be manipulated so the drug solution is at the top of the syringe and the air bubble sits between the syringe plunger and the drug solution. During administration, the air bubble flushes the drug solution through the MAD, meaning that air is retained within the dead space rather than drug solution.



Drug solution Air bubble

- A short sharp expulsion of drug from the syringe ensures that the liquid is optimally atomized.

- The nostrils should also be examined for either excess mucus or blood as this will hinder drug absorption; nostrils may also be suctioned prior to administration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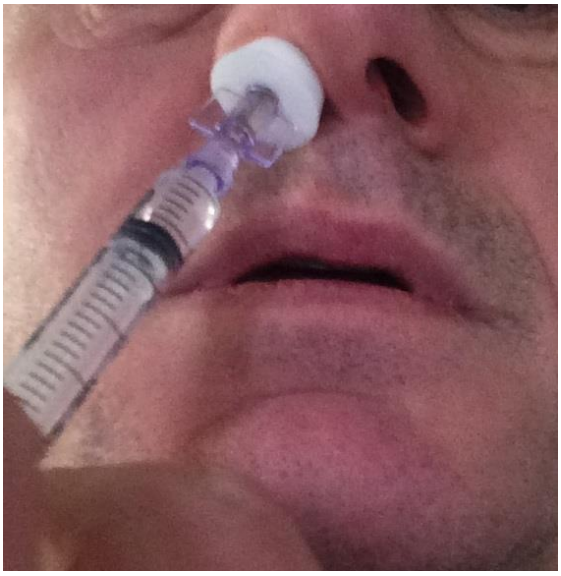

5. MONITORING COMPLIANCE WITH THIS GUIDELINE

5.1 Standards/Key Performance Indicators

- Adequate sedation and compliance for anaesthesia in children or adults with learning disabilities
- Improved pre-operative sedation in the target groups as reported by the Anaesthetic team

5.2 Process for Implementation and Monitoring Compliance and Effectiveness

- Guideline will be presented at anaesthetic governance and distributed to relevant parties.
- Guidelines will be available on the intranet
- Feedback will be sought from relevant parties and Datixes reviewed.

6. ASSOCIATED CLINICAL GUIDELINES OR POLICIES/PROCEDURES

- RDUH Medicines Management Policy 2024 https://royaldevonstaff.nhs.uk/download.cfm?ver=43285
- RDUH Mental Capacity Act Policy 2025 https://royaldevonstaff.nhs.uk/download.cfm?ver=24599
- Consent for Examination or Treatment Policy. RDUH. 2021. https://royaldevonstaff.nhs.uk/download.cfm?ver=21896

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